

Gamified ADHD Interventions through Human Computer Interaction: A Thorough Study of BCI, AR and Cognitive Training Games for Children with ADHD

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Abstract

In this semi-systematic literature review, a detailed study of the role of Human-Computer Interaction (HCI) in creating game-based solutions for Attention-Deficit/Hyperactivity Disorder (ADHD) among children is conducted. Six peer-reviewed research studies were selected. The study demonstrates that HCI can serve as a major therapeutic mechanism by transforming digital platform-based cognitive training into engaging, interactive experiences. These approaches not only improve focus but also enhance the overall effectiveness of interventions. Key findings from the analyzed studies are discussed, and future research directions are proposed, including multimodal hybrid systems with adaptive personalization and accessibility features to further improve outcomes for children with ADHD.

Keywords: ADHD Treatment; Augmented Reality; BCI; Children with ADHD; Cognitive Training; Gamification; HCI; Serious Games

1 Introduction

Attention Deficit/Hyperactivity Disorder (ADHD) is a very common neurodevelopmental disorder among children, and the symptom of ADHD includes inattention (as they cannot focus easily), hyperactivity (continuous movement that is not normal), and impulsivity (acts without thinking anything) that directly impacts children's academic performance, socialization, daily functioning and emotional stability. ADHD is a serious chronic and enervating disorder that impacts a person's life in many ways that create obstruction in their growth. Common treatments of ADHD are psychostimulant medicine, behavioral therapy, cognitive training and so on. But there are several side effects of those treatments, such as medication based treatments can lead to only short term improvements, behavioral therapy can make a child dependent on adults and so on. On the flip side of the coin, Human Computer Interaction (HCI) can be a very good alternative to traditional interventions that are used to treat ADHD among

children. As HCI focuses on developing interactive technology that ensures intuitive, efficient and comfortable experiences by pointing out users needs, so solutions developed keeping in mind the psychological needs of children with ADHD. Using interactive technology it is possible to develop solutions that are focused on cognitive and behavioral training and the interventions can be more engaging, accessible and adaptive to its user. In the sector of ADHD treatment research, HCI is progressing rapidly in several fields, such as –

- Serious games: focuses on cognitive and executive functioning.
- Brain Computer Interaction (BCI): uses neurofeedback as input data for game controls.
- Augmented Reality (AR) and Natural User Interfaces: Focuses on cognitive training.
- Interactive Assessment Tools: uses gamified solutions to transform clinical observations more effective and realistic.

Those approaches are very constructive in incorporating sensory, cognitive and neurophysiological elements into game frameworks, that are expected to be more potent to young children with ADHD due to their nature being interested in computer-based solutions. The main focus of this study was to investigate the impact of the HCI based solutions in order to develop awareness, executive functioning, focus, and control among the children with ADHD in order to increase their self-esteem and help them grow. Instead of technical intervention based review, this study focuses on extensive theoretical connections between the ideas, methodological trade-offs and idea transferability, and ecological validity.

2 Research Question

Though recent HCI based interventions are showing new possibilities to treat ADHD among children that is beyond traditional clinical and educational approaches, but mastery of their design principles and cognitive mechanism is very important to correlate their effects across different research studies and solutions. Therefore, to guide the literature review we came up with following research questions-

- To what extent HCI and Gamification based approaches, such as Serious Games, BCI and AR environments, help to develop cognitive and behavioral interventions for the children with ADHD?
- What are the common design patterns, their limitations and future possibilities based on recent studies?

Using those questions we tried to figure out not only their effectiveness, but also ways that interactive systems developed cognitive theory, user engagement and expected outcomes. Through a detailed cross-paper study we tried to find their relationships, current state of knowledge and limitations.

3 Methodology

To identify and narrow down the interconnection of HCI and gamified ADHD interventions a semi-systematic literature review was coordinated. This study was designed such as way that it can be replicated and transparent at the same time. The main focus of this search was to find six peer-reviewed articles from reputed conference and journals that align with the topic and the idea intersects to our research ideas. Table 1 below presents a brief idea about the sources that was used to find the papers, keywords that were used to search and the outcome of those searches.

Table 1: Literature search - Source, Keywords and Papers.

Source	Keywords	Papers
https://scholar.google.com/	gamified app to treatment ADHD children	[21, 3, 9, 18, 17, 6, 2, 7]
https://scholar.google.com/	ADHD digital treatment	[6, 12]
https://untexas.summon.serialssolutions.com/	Gamification, ADHD Children	[9, 20, 5, 8]
https://ieeexplore.ieee.org/	ADHD children serious games	[4, 10, 8, 15, 11, 18]
https://ieeexplore.ieee.org/	BCI based games for ADHD treatment	[19, 1, 23]
https://ieeexplore.ieee.org/	BCI based games for attention training	[16, 13, 22, 14]

All the papers here were published between 2010 to 2025 in order to cover a decade of technological evolution in HCI. The database used for the search were mostly Google Scholar, IEEE Xplore, UNT Library Resources to find articles, MDPI and SpringerLink. Based on this targeted search six papers were selected. For the paper selection some selection criteria were used. Those selection criteria were –

- Peer reviewed conference or journal papers.
- Gamified app or games that involved case study to treat ADHD among children.
- Focused discussion on HCI topics such as design principles, developed interface technology and interaction modality.
- Presents empirical data or a detailed design framework.

Though many of the paper were fulfilling the initial selection criteria, therefore I had to set some exclusion criteria including the comparison of the paper to my primary research topic. Those exclusion criteria were –

- Non-game, gamified or non-interactive applications.
- Review based papers.
- Papers that don't have case studies.

- Papers that have case studies, but participant were not between 6 to 12 years old.
- Papers that only focuses on diagnosis.

So the selected papers, that is - [4], [5], [6], [8], [11] and [13], were the most relevant ones and close to the core research idea. However, all the papers were not on the same platform or topics, as on the same topic or platform there were not many papers available. Some closely matched papers did not fulfill the selection criteria, and because of that they were excluded from the list.

4 Results

This literature study presents that HCI is one of the core component in threptic practices which we can call digital therapy and at the same time, it is dynamic and can be very effective in some cases. This study was divided into four concepts or themes in order to understand the ideas better and demonstrate how those ideas are interconnected, their differences, and how the ideas are evolving over time. In the next part of the study we will discuss the concepts in details.

4.1 Concept 1: Considering engagement as the major mechanism for therapy

According to all the six studies that were reviews presents strong evidence that engagement not only worked to treat ADHD among children, but also acted as one of the major mechanisms for therapeutic effectiveness. In general children with ADHD lacks motivation to do repetitive work and losses focus very fast if they find no interest in the task. However, interactive solutions creates a cognitive motivational loop that involves rewards or task achievement, which mitigate the issue with lack of motivation and interest. Therefore, serous games like Antonyms [8] and BRIGHT [5] focused on the modification of the motivational structure of cognitive tasks. Those systems decreased the emotional unwillingness and uplifted the determination among young children by lodging restraint from the portrait of a superhero or providing constant positive feedback and personalized positive reinforcement. The focus of those systems were keeping the game environment challenging but not overwhelming through a constructive and supportive game mechanism. It was a very crucial part of the system as it helped the children to stay motivated and maintain their engagement to the game. Those mechanisms played a significant role to the systems as the game performance were highly dependent on the task novelty and understanding interests.

On the other hand, systems like ATHYNOS [4] and BRAVO [6] uses physical interaction, such as full body movement and area exploration in order to increase affective engagement. Based on the embodied cognition theory, physical activity based approaches can active neural networks in our brain by increasing brain activity that support attention and regulation. Because of the gamified approach and socially acceptable nature of the system those physical activities can be very

effective to treat ADHD and it feels more like a game than any clinical therapy and it can directly create significant impact on motivational deficits.

BCI based systems can also create positive feedback as games like [13] uses neural response and children can actually see their own brain activity in the form of attention level while controlling the game. Such gamified systems increases curiosity and internal motivation among children resulting improvement in their attention level. This has a significant difference with other motor based systems that involves physical activity through an interface, and this system completely depends on the mental state of the user.

Synthesis: Considering all the scenarios it can be stated that physical or mental engagement is the catalyst that makes cognitive training possible and effective. Through this engagement procedure, those HCI-based systems create a loop of interactive challenges spiking motivation, leading to sustained attention, improved performance and self-control.

4.2 Concept 2: Integrating the cognitive mechanisms into digital interaction design

The next concept found from the studies is transforming abstract cognitive constructs such as, inhibition, attention and working memory into sensible and interactive mechanisms, which is one of the major contribution of HCI to ADHD treatment of children. If we consider ANTONYMS [8] as an example, it is a serious game constructed based on neuropsychological Dual Pathways Model of ADHD by converting unease tasks like Stroop and Flanker tests into interactive gameplay. In the game, the Woodland scenario explains that the player must control themselves to move the most accessible leaf first in order to solve it and must come up with some other strategy and planning. In this test the child faces discomfort as a narrative conflict, instead of an artificial clinical task. This game strategy has significant similarity with standard clinical approaches that provides this HCI based approach an ecological validity.

On the other hand, EMOGALAXY [11] uses similar approach in order to test a ranking of emotional conditions such as, recognition, expression and regulation. Its camera based setup monitors users emotional condition to send input to the game mechanism (Ladder and Snake) that require emotional regulation. This leads to emotional self-regulation into an interactive decision-making process and leads towards developing social skills. However, the BCI-Based 3D Game [13] handles controlling attention in a different way, and it uses EEG based data to control the movement of a virtual hand. To succeed in this game the user must gain high control on their attention to get control over the game. This approach uses mental state of the user into controllable game element which is an unique feature of HCI that converts the traditional passive experience of a continuous performance test into an active and inclusive digital activity.

In an AR based system like ATHYNOS [4] executive functioning gets converted into physical positioning and area-based problem solving. This approach presents that the mental work of our movement and perception is strongly connected to our way of thinking. This approach suggests that reduction in time

to complete a task compared to traditional games proves this game mechanism helps to increase the efficiency of user's mental energy or brain activity.

Synthesis: Those study proves that HCI is transforming unobservable cognitive mechanism into visible and controllable interactive structure and this is one of the major achievement of HCI. Using this game mechanics it is possible to motivate children to practice self-control, attention and emotional regulation. Because of the gamified nature of those systems children tend to be more motivated than any traditional clinical approach.

4.3 Concept 3: Transformation of ADHD training method from traditional to HCI based neuro-embodied systems

The reviewed papers presents a good understanding of the development of HCI based technology over the years to treat ADHD among children, where it started from traditional computer based systems to advanced AR and BCI based technology that is advancing rapidly and opening doors of hope to completely treat ADHD in early and moderate stage. Gamified approaches such as, BRAVO [6] shows that traditional computer based approaches can improve the mental condition of ADHD children. However, the impact is moderate and have limited engagement and lack of motivation in those base level approaches, and can be improved farther to get better results to treat ADHD. On the other hand, games like ANTONYMS [8] and BRIGHT [5] shows significant shift from that stage by incorporating rewards based systems that includes narrative framing, adaptive levels and psychologically advanced environments, which turned the systems from “training approach” to “engagement based approach”.

Systems like ATHYNOS [4] and components from BRAVO [6] introduces physical involved physical and movement based activity through technology like the Kinect that engages children to physical activity and improve their engagement in order to improve their motor skills or controls and spatial reasoning. On the flip side of the coin, BCI based platforms [13] focuses on the mental stability and control among children. This type of technology depends on psychological measurement, dynamic difficulty adjustment and calibration based on the users brain activity. Moreover, the BRAVO [6] project predicted at which direction technology was advancing integration AR, VR, neurofeedback and affective agents as a cross platform multimodal therapeutic ecosystem.

Synthesis: This progress shows that how HCI is advancing towards treating ADHD from a screen based training method to multimodal interventions that involves engagement and leads to a more motivating and effective environment. Making the whole process interesting for children making them cognitively, emotionally, physically and psychologically stable.

4.4 Concept 4: Limitations in different systems and their common methodological challenges

Though all the six studies here are showing good progress, still they have a few structurel limitations that are very important to point out. All of them only

worked on a very small sample size. For example, for ANTONYMS project they had a group of 8 ADHD children aged between 8 to 11 years old, ATHYNOS had 11 participants with ADHD aged between 7 to 10 years old and BCI based game had 10 participants aged between 6 to 12 years old. All of them had only a few participants that can be used for a pilot study, but it is difficult to predict what would happen if it were used by a large group of participants.

None of those studies lasted more than a few weeks and no long-term follow-ups were conducted to help us to understand any improvements. More importantly, there is no study of differences between lab-based improvements and real-life improvements. Besides, those outcomes are unclear as those were not a controlled group experiment and no non-technical therapeutic activity was performed to get a hit about if those results were because of the extra attention the participants were getting or they were involved in gamified environments, rather than actually improving. Moreover, there are a few technical barriers in the implementation in some of those systems. One of the major drawbacks of BCI-based system is they suffer from significant amounts of noise and electrodes sometimes fail to get appropriate data. Some people are sensitive to physical touch and refuse to use wearable devices. AR systems can be expensive to setup and requires special hardware. On the other hand, motivating children to use tools like BRIGHT and EMOGALAXY can be challenging sometimes as they may lose interest very fast. Those drawbacks create significant obstacles to using advanced solutions like those.

5 Discussion

This study reveals that HCI is fundamentally reshaping the ADHD treatment by creating engaging, organized and theoretically grounded digital solutions. Serious games, BCI and AR based software can be a very good alternative of clinical therapy and can be merged into one solution as well. Serious games can be a good solution for inhibitory control, BCI incorporates real time neural feedback loops and AR based solutions can be useful for executive functioning. A very common topic among all those systems is their tightly coupled feedback loops as their main mechanism. Whether the feedback is based on performance or phycology or affect, this adaptation plays a critical role for constructing self-regulation and balancing engagement. One of the bright sides of those solutions is that each one of them are following any specific cognitive theory. For example, ANTONYMS is based on Dual Pathway Model, BRIGHT following Malone’s theory of intrinsic motivation, BRAVO is based on self-regulation and social cognitive theory. This theoretical alignment verifies that those solutions are not just for fun, instead they have clear goals to achieve. One more major benefit of HCI based systems is that they can mitigate the boundary between assessment and rehabilitation. ANTONYMS presents connections between in-game error and clinical measurements, which makes it an embedded assessment tool. BCI based games can provide real time data on neural feedback, which will provide valuable insight into children’s progress.

6 Future Research Direction

Though this study provides evidence that the study on ADHD treatment is progressing rapidly, and to get the best results it is important to test those prototypes in larger group of people in real world scenarios. Long-term observation should also be a very crucial part of this study to ensure meaningful outcomes. Constructing hybrid systems incorporating BCI, AR and Serious Game can be a better and more effective option to treat ADHD. AI based personalization should be used to adjust the difficulty level, detect needs and identify appropriate game mechanics for each individual. To support clinical adaptation, systems must integrate workflows through dashboards and convert game based data into actionable insights.

7 Conclusion

This study combined the six targeted research papers in order to get deep insights into the recent advancements of gamified HCI technology to treat ADHD among children. The analysis identified four broad concepts to get a better understanding of the solutions and their workflow. Also, it has become clear that HCI and Gamification has great potential to treat ADHD among children due to its interactive nature that focuses on physical, psychological and cognitive development, and multimodal based approaches may create significant impact. Though actual clinical study-based comparison is missing, it should be incorporated into the future solutions.

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